

The quality of life of children and adolescents with ADHD undergoing outpatient psychiatric treatment: simple disorders of activity and attention and hyperkinetic conduct disorders in comparison with each other and with other diagnostic groups

Helmut Remschmidt · Fritz Mattejat

Received: 27 September 2010 / Accepted: 5 October 2010 / Published online: 10 November 2010
© Springer-Verlag 2010

Abstract (1) How does the quality of life of patients with ADHD treated in an ambulatory care setting compare to that of other patient groups in child and adolescent psychiatry? (2) Can differences in the quality of life be demonstrated between patients with simple disorders of activity and attention and those with hyperkinetic conduct disorders? (3) How does the quality of life in these patient groups change over one year of treatment? The Inventory for the Assessment of Life Quality in Children and Adolescents (*Inventar zur Untersuchung der Lebensqualität von Kindern und Jugendlichen*, ILK) was applied to a sample of 726 patients derived from nine different outpatient practices for child and adolescent psychiatry. Among them were 196 patients with a simple disorder of activity and attention and 64 with a hyperkinetic conduct disorder. A comparison between these two groups was the main aim of the study. The mean age of the patients in the sample (all diagnoses) was 8.7 ± 3 years. The two groups of hyperkinetic patients made up 35% of the overall sample, and both of them showed a marked male predominance. The hyperkinetic patients tended to have lower quality-of-life scores than patients in the other diagnostic groups. Longitudinal observation revealed improvements in the quality of life across all patient groups, but the patients with hyperkinetic disorders (both groups) improved the least. The parents of the hyperkinetic patients, too, reported suffering greater stress because of their children's condition than the parents of children with other types of disorders. The ILK instrument has test-metrical qualities that render it usable and capable of holding its own among other, comparable instruments. It can be used to assess the

quality of life of children with various diagnoses. Children with ADHD tend to have the least favorable quality-of-life scores, yet they do show some degree of improvement in their quality of life after a year of treatment.

Keywords ADHD · Quality of life · Naturalistic · Treatment conditions · Cross-sectional and longitudinal study

Introduction

It has become generally accepted in recent years that evaluations of disease course and of treatment success in child and adolescent psychiatry must be based not only on disease-related and disease-specific variables, such as clinical manifestations, psychotherapeutic methods, and medications but also on variables reflecting the patients' social situations, integration into their families, schools, occupations, and relationships to peer groups. Such factors are often subsumed under the general heading of "quality of life." The quality-of-life concept is particularly important for patients with chronic diseases, as well as for persons who have had to undergo extensive surgery or other medical treatments that can cause major, unavoidable side effects as a necessary accompaniment to the achievement of a defined therapeutic goal. In considering the possible indications for such interventions, physicians often ask what quality of life the patient can be expected to have after the procedure or treatment in question.

The specialty of child and adolescent psychiatry often deals with chronic diseases. Here, too, it has become generally accepted that treatments should be evaluated with respect to their effects on the quality of life.

H. Remschmidt (✉) · F. Mattejat
Universität Marburg, Marburg, Germany
e-mail: remschm@med.uni-marburg.de

Quality of life has been defined as “an individual’s perception of his position in life, in the context of the culture and value systems in which he lives and in relation to his goals, expectations, standards and concerns” (WHO QoL Group 1993, 1995).

Among all of the diseases that are dealt with in child and adolescent psychiatry, attention-deficit/hyperactivity disorder (ADHD) is one to which considerations of the quality of life are particularly relevant. Because of its high prevalence, its persistence over time, and its social-psychiatric effects on the family, on schooling, and on the occupational development of the affected children and adolescents, it represents a major personal handicap and is also associated with high costs of treatment and care (Swensen et al. 2003).

This fact was certainly what motivated the journal ECAP’s editors to publish, in 2006, a supplementary issue devoted to the subject of “Attention-Deficit/Hyperactivity Disorder: Observational Research in Europe” (ADORE) (Rothenberger et al. 2006), which is included in this article’s References. The Supplement contained an initial report of findings from a pan-European ADORE study, in which parents of children with ADHD from 10 European countries were asked to assess their children’s quality of life by means of a questionnaire, the Parent Report Form of the Child Health and Illness Profile—Child Edition (CHIP-CE). In this initial, cross-sectional study, the CHIP-CE was found to be a valid and reliable instrument for assessing the quality of life of children with ADHD (Riley et al. 2006a).

In a further study published in the same Supplement (Riley et al. 2006b), the authors determined the factors that were associated with the patients’ health-related quality of life (HRQoL). In addition to the hallmark manifestations of ADHD, these factors included conduct disturbances, problems with peer relationships, bronchial asthma, a variety of other bodily symptoms, and impaired motor coordination. The relevant familial factors included physical or mental illness in a parent (of which the child’s disease might be a contributing cause), being raised by a single parent, and maternal cigarette smoking during pregnancy. All of these factors were associated with a low health-related quality of life. Like first study, this one was conducted as a baseline study and contained no longitudinal data. The same was true of the third study in the Supplement, in which the parents of children with ADHD were asked to fill out a questionnaire with data that were used to calculate a Family Strain Index (FSI). In the first step of this study, satisfactory test-metric data were obtained for the particular instrument that was used; as a result, the authors recommended using it to assess the patients’ quality of life as well. This study, like the other two, was carried out in the ADORE network. The questionnaire was completed by parents from a total of 1,478 families in 10 European countries. All three of these studies were mainly intended to

contribute to the development of satisfactory investigational methods (Riley et al. 2006c).

Only a few studies to date have addressed the quality of life of children and adolescents undergoing psychiatric treatment, and even fewer have taken account of the longitudinal aspect. In this article, we report the findings of a study of this type.

Questions

Our study addressed the following questions:

1. How does the quality of life of patients with attention-deficit/hyperactivity disorder–hyperkinetic syndrome (ADHD/HKS) undergoing outpatient psychiatric treatment compare to that of patients with other diagnoses?
2. Does the quality of life of patients with simple disorders of activity and attention differ from that of patients with hyperkinetic conduct disorders?
3. How does the quality of life change over the course of outpatient treatment (the longitudinal aspect) in patients with simple ADHD, and in those with ADHD and conduct disorders, in comparison with patients with other diagnoses?

The disorders were classified according to the criteria of the German edition of the ICD-10 (Dilling et al. 1997). The terms ADHD and HKS are used synonymously.

Inventory for the assessment of life quality in children and adolescents (*Instrument zur Erfassung der Lebensqualität von Kindern, ILK*)

The basic components of the quality-of-life concept are listed in Fig. 1; they include influencing factors, objective aspects, and subjective aspects. The instrument that we have developed, the Inventory for the Assessment of Life Quality in Children and Adolescents (ILK) (Mattejat and Remschmidt 2006), is based on subjective judgments of the quality of life made by the affected children and adolescents and by their parents and psychotherapists.

The rating forms of the ILK are detailed in Table 1.

The different domains of the ILK rating scales for children and adolescents are shown in Table 2. These involve seven areas (family, school, physical health, etc.), with two additional domains for the patients and their parents and two further ones that are addressed solely to the parents.

Data on the reliability and validity of the ILK are shown in Table 3. The data indicate that this instrument’s reliability and validity are satisfactory. Correlations between the ILK and other instruments used to assess children’s quality of life (Kindl and DIKJ) are indicated in the last

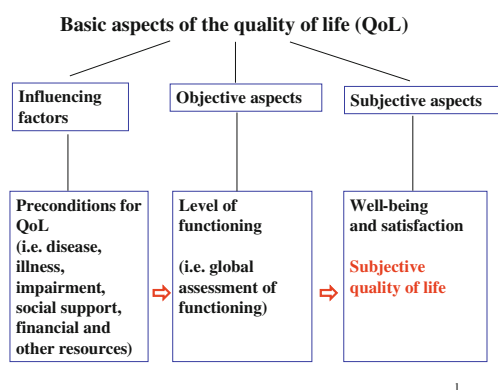


Fig. 1 Basic aspects of the quality of life (QoL)

two lines of the Table (Bullinger et al. 1994; Stiensmeier-Pelster et al. 2000).

A number of studies on the quality of life of patients in child and adolescent psychiatry have now been performed with the aid of the instrument that we developed; we cannot discuss these studies in detail here. The study on which we report in this article concerns not just patients with ADHD

but also patients with a number of other disorders from the spectrum of problems that are seen in child and adolescent psychiatry.

Description of the sample population

We employed a sample population that was obtained in the setting of the KJP Quality Project. This project involved a large, unselected sample of patients who presented consecutively (without gaps) to 9 outpatient practices for child and adolescent psychiatry. The subjects were studied longitudinally at three time points: T1, presentation to the outpatient practice; T2, three months after presentation; and T3, one year after presentation. Data were collected from the psychotherapists (e.g., for basic personal reference data) and from the subjects' main reference persons (typically, parents) at all three time points, by telephone interview and by questionnaire. This study's main findings are conveyed in a final report (Mattejat et al. 2006), which also contains an extensive description of the study sample.

Table 1 ILK parallel rating forms

Source of information	Aspects	Instrument
Children (6–11)	Self-reporting of QoL on 9 rating scales	Child interview with rating forms
Adolescents (12–21)	Self-reporting of QoL on 9 rating scales	Adolescent rating form
Parents (mother/father)	Rating of the child or adolescent's QoL on 9 rating scales Proxy measure; additional assessment of the burden and impact on the parents that are caused by the child's illness, diagnostic evaluation, and treatment	Parent rating form
Psychotherapists	Level of functioning; 9 rating scales	Therapist rating form

Table 2 Rating scales of the ILK: different domains

(1) School
(2) Family
(3) Other children (social contacts with peers)
(4) Alone (interests and activities when alone; ability to occupy him- or herself)
(5) Physical health ("health")
(6) Psychological well-being ("nerves, mood")
(7) Global rating ("all together")
Additional scales for patients and their parents:
(8) Problems: negative impact/stress associated with the present disorder/disease
(9) Evaluation/treatment: negative impact/stress associated with diagnostic evaluation and treatment
Additional scales only for parents:
(10) Stress/negative impact on the parents associated with the child's present disorder/disease
(11) Stress/negative impact on the parents associated with the child's diagnostic evaluation and treatment

Sum of scores on scales (1)–(7): QoL score

Table 3 The reliability and validity of QoL scores from the ILK

Internal consistency of QoL scores in general population samples: 0.63 (children) to 0.76 (parents)
Internal consistency of QoL scores in patient samples: 0.53 (children) to 0.72 (parents)
Retest correlation at 5 weeks: 0.72 (children)
Retest correlation at 3 weeks: 0.80 (parents)
Retest correlation at 6 weeks: 0.68 (parents)
Parent–child correlation: 0.53
Correlation with KINDL®: 0.65 (children)
Correlation with DIKJ: 0.64 (children)

A total of 1029 patients were included in the study. The point of departure for the present study was the “working sample” used in the KJP Quality Project, which consisted of the patients for whom a complete data set was obtained, i.e., for whom at least the following were present:

- a diagnosis made by the leading child psychiatrist caring for the patient, and
- all three telephone interviews for all three time points (T1, T2, and T3).

This working sample (reduced sample) consisted of 728 patients (cf. the detailed project report, Mattejat et al. 2006, p. 93) and was further reduced to 726 patients because of missing values for two patients.

The distribution of diagnoses in the working sample is shown in Table 4.

As seen in the Table, 196 patients had a simple disorder of activity and attention, while 64 of the children and adolescents had been given the diagnosis of a hyperkinetic conduct disorder. All-in-all, the hyperkinetic disorders accounted for the largest diagnostic group by far, with 260 patients and thus more than 35% of the entire sample. Next in frequency were the specific emotional disorders of childhood, adaptive disorders, and conduct disorders.

The five most common diagnoses listed in the Table accounted for more than three-quarters of the entire sample. The remaining 180 patients were distributed over a wide variety of other diagnostic groups.

Treatment

Treatment was provided in accordance with the guidelines of the three children and adolescent psychiatric organizations (Deutsche Gesellschaft für Kinder- und Jugendpsychiatrie et al. 2007) and consisted of the following three components, in varying combinations:

1. education of the patients and their parents about the nature of the problem (psychoeducative approach),
2. behavior-oriented psychotherapeutic measures, and
3. prescription of medication (generally, either a stimulant or atomoxetine).

In the evaluation below, we will consider the quality of life of patients with

- simple disorders of activity and attention and
- hyperkinetic conduct disorders

and compare it with the quality of life of patients in the three other major diagnostic groups (emotional disorders of childhood, adaptive disorders, and conduct disorders). In all of the evaluations, we will thus make use only of the sample of patients with the five most common diagnoses (524 patients in all), in order to assess the degree of impairment of quality of life, if any, in the two types of hyperkinetic disorder in comparison with the other three diagnostic groups.

The age distribution in the sample is shown in Table 5, and the sex distribution in Table 6. Table 7 provides an

Table 4 Distribution of diagnoses in the working sample of the KJP quality project

	N	%
Adaptive disorders (F 43)	96	13.2 ^a
Simple disorder of activity and attention (F90.0, 90.8, 90.9)	196	27.0 ^a
Hyperkinetic conduct disorder (F90.1)	64	8.8 ^a
Conduct disorder (F91, F92)	67	9.2 ^a
Specific emotional disorder of childhood (F93)	123	16.9 ^a
Other diagnoses	180	24.8
Total	726	100%

^a Sample with the five most common diagnoses: $n = 524$ (75.2% of the overall sample)

Table 5 Age distribution in the sample

	Mean	Standard deviation	Number of subjects
Adaptive disorders (F43)	9.33	3.42	96
Simple disorder of activity and attention (F90.0, 90.8, 90.9)	8.16	2.42	196
Hyperkinetic conduct disorder (F90.1)	9.25	2.98	64
Conduct disorder (F91, F92)	8.46	3.66	67
Specific emotional disorder of childhood (F93)	8.78	2.93	123
Other diagnoses	8.67	2.99	546
Results of analysis of variance: $df = 4$, $F = 3.41$, $P = 0.009$			

Table 6 Sex distribution in the sample

	Female <i>N</i> (row %)	Male <i>N</i> (row %)	Overall <i>N</i> (column %)
Adaptive disorders (F43)	42 (44)	54 (56)	96 (18)
Simple disorder of activity and attention (F90.0, 90.8, 90.9)	55 (28)	141 (72)	196 (36)
Hyperkinetic conduct disorder (F90.1)	14 (22)	50 (78)	64 (12)
Conduct disorder (F91, F92)	18 (27)	49 (73)	67 (12)
Specific emotional disorder of childhood (F93)	41 (33)	82 (67)	123 (23)
Other diagnoses	170 (31)	376 (69)	546 (100)
Pearson- $\chi^2 = 11.395$, $df = 4$, $P = 0.022$			

Table 7 Family settings of patients in the sample

	Both father and mother in household* <i>N</i> (row %)	Single parent <i>N</i> (row %)	Other constellation <i>N</i> (row %)
Adaptive disorders (F43)	68 (71)	27 (28)	1 (1)
Simple disorder of activity and attention (F90.0, 90.8, 90.9)	172 (88)	22 (11)	2 (1)
Hyperkinetic conduct disorder (F90.1)	51 (80)	12 (18)	1 (2)
Conduct disorder (F91, F92)	50 (75)	16 (23)	1 (2)
Specific emotional disorder of childhood (F 93)	104 (85)	19 (25)	0 (0)
Other diagnoses	445 (82)	96 (17)	5 (1)
Pearson- $\chi^2 = 16.993$, $df = 8$, $P = 0.030$			

* The father and mother need not be the child's biological parents; the deciding factor in this statistic is that there are two parents jointly raising the child

overview of the patients' family settings, broken down by diagnostic group.

As seen in Table 5, the peak ages in the sample are in the primary school years. As seen in Table 7, the sample consists of about 70% boys and 30% girls. This age and sex distribution is typical for outpatient samples in child and adolescent psychiatry.

The male predominance is seen to be particularly pronounced among children with hyperkinetic disorders and conduct disorders. It is noteworthy, too, that children with hyperkinetic conduct disorders (F90.1) are, on the average, markedly older than those with simple disorders of activity and attention (F90.0, 90.8, 90.9). This can be taken as an indication that children who present at a later time (i.e., when they are older) are more likely to have a hyperkinetic conduct disorder; in other words, if the patient presents

later, there is a greater danger that the disorder will have progressed to include, not just a disturbance of activity and attention but also a disturbance of conduct. On the other hand, children who present at younger ages are more likely to have a simple ADHD. If this interpretation of the figures presented here is correct, then one would also expect treatment to succeed less often in patients with hyperkinetic conduct disorders (who are more likely to have undergone chronification, with more varied and severe manifestations) than in patients with simple disorders of activity and attention. This hypothesis can be tested with data from the sample used in the present study.

As seen in Table 7, most (82%) of the children and adolescents lived with two parents in the same household. Single parents (usually single mothers) were particularly frequent among patients with adaptive disorders. It is

Table 8 Direct assessment of the success of treatment: response of the patient's main reference person to the question whether, and how, his or her problem has changed

	Fully improved	Markedly improved	Somewhat improved	Unchanged	Worse	Overall
Adaptive disorders (F43)	15 16.0%	37 39.4%	19 20.2%	17 18.1%	6 6.4%	94 100.0%
Simple disorder of activity and attention (F90.0, 90.8, 90.9)	14 7.3%	94 48.7%	52 26.9%	22 11.4%	11 5.7%	193 100.0%
Hyperkinetic conduct disorder (F90.1)	2 3.2%	32 51.6%	15 24.2%	8 12.9%	5 8.1%	62 100.0%
Conduct disorder (F91, F92)	7 10.8%	31 47.7%	13 20.0%	10 15.4%	4 6.2%	65 100.0%
Specific emotional disorder of childhood (F93)	14 11.6%	59 48.8%	27 22.3%	20 16.5%	1 0.8%	121 100.0%
Total (sum)	52 9.7%	253 47.3%	126 23.6%	77 14.4%	27 5.0%	535 100.0%
Pearson- $\chi^2 = 20.278$, $df = 16$, $P = 0.208$						

Bold numbers indicate the differences (trends) between the two ADHD groups

highly noteworthy that patients with simple disorders of activity and attention were the most likely of all (88%) to live in intact families with two parents. With respect to the distribution of socioeconomic status (not shown in any of the tables), we will simply state here that there were no significant differences among the diagnostic groups (for more extensive information, see Mattejat et al. 2006).

Results

Direct assessment of the success of treatment and of the change in overall well-being

Before considering the quality of life of the patients in our sample and its development over the course of a year of treatment, we would like to take a look at the answers that the patients' main reference persons gave, one year after presentation, to questions concerning

- the success of treatment and
- the overall development of the child or adolescent.

The data presented in Table 8 reveal, first of all, that there were no statistically significant differences in the rates of treatment success in the five diagnostic groups. In all groups, improvement was the most common outcome, observed in about 80% of patients. This, of course, implies that some 20% of patients were no better after treatment than before, in the estimation of their main reference persons.

It is also worth noting that children and adolescents with ADHD were much less likely to be judged as "fully improved" one year after presentation than those in the other diagnostic groups. Thus, ADHD, unlike the other diagnoses, seems to be a chronic disorder in which a total remission is rarely, if ever, achievable.

On the other hand, when asked about the patients' overall well-being, the reference persons answered "very good" for a relatively high percentage of the patients in both ADHD groups.

Thus, even though only a small percentage of patients with ADHD attain complete or near-complete remission, many are nonetheless reported to experience a major improvement of their overall well-being, as shown in Table 9.

In general, the course over a year of treatment was positive: in all of the diagnostic groups, treatment was accompanied by an improvement of the patients' problems as well as of their overall well-being. Full improvement was relatively rare among patients with ADHD, yet these patients, too, often felt markedly better overall than they had felt at the start of treatment.

This generally positive picture, however, requires some qualification. About 15 to 20% of the patients were no better at one year than they had been on presentation. This group of patients, for whom the therapeutic interventions were unsuccessful, should be analyzed further with a view toward finding possible ways of improving their care.

More extensive and detailed analyses of the effectiveness of the psychiatric treatments for children and adolescents that are considered here can be found in Bachmann et al. (2010).

Before-and-after comparison of quality of life

Quality-of-life score

The means and standard deviations of the quality-of-life scores, LQ₀₋₂₈, for the individual diagnostic groups, both at baseline (initial outpatient presentation, time point T1) and one year later (time point T3), are shown in Table 10 along

Table 9 Changes in overall well-being: response of the patient's main reference person to the question, "How is your child's overall well-being compared to one year ago?"

	Very good	Good	So-So	Poor	Very poor	Overall
Adaptive disorders (F43)	33 34.4%	50 52.1%	10 10.4%	3 3.1%	0 0.0%	96 100.0%
Simple disorder of activity and attention (F90.0, 90.8, 90.9)	59 30.1%	107 54.6%	22 11.2%	8 4.1%	0 0.0%	196 100.0%
Hyperkinetic conduct disorder (F90.1)	17 26.6%	32 50.0%	10 15.6%	4 6.3%	1 1.6%	64 100.0%
Conduct disorder (F91, F92)	15 22.4%	42 62.7%	8 11.9%	2 3.0%	0 0.0%	67 100.0%
Specific emotional disorder of childhood (F93)	36 29.3%	67 54.5%	17 13.8%	3 2.4%	0 0.0%	123 100.0%
Total (sum)	160 29.3%	298 54.6%	67 12.3%	20 3.7%	1 0.2%	546 100.0%
Pearson- $\chi^2 = 14,040$, $df = 16$, $P = 0.596$						

Bold numbers indicate the differences (trends) between the two ADHD groups

with the calculated pre/post effect strengths and the results of the corresponding analysis of variance.

The quality of life score LQ_{0-28} can range from 0 (least favorable value) to 28 (most favorable value). All of the values reported here lie, as expected, below the mean for the general population; that is, children and adolescents who are in psychiatric care have a markedly impaired quality of life compared to the population at large. We have already reported findings of this type before (Mattejat and Remschmidt 2006).

We will not deal any further here with the fundamental question of the general impairment of quality of life among mentally ill children and adolescents. Rather, we will consider the differences between patients in the individual diagnostic groups, and the results obtained in each group after one year of treatment.

Table 10 and the corresponding analysis of variance make it clear that:

- The individual diagnostic groups differ from one another markedly, and to a high degree of statistical significance, in the quality of life. Patients in the two ADHD groups had the worst quality of life of all groups, both on initial presentation and one year later. The impairment of quality of life was particularly severe in patients with hyperkinetic conduct disorders.
- The overall sample still showed highly significant improvement after one year of treatment.
- There was no significant interaction between the factors "diagnosis" and "time point," i.e., comparable degrees of improvement were found in all diagnostic groups. On the purely descriptive level, however, the calculated effect strengths indicate that more marked improvement was obtained by patients with specific emotional disorders and conduct disorders than by patients in either of the two ADHD groups.

Table 10 The quality-of-life scores LQ_{0-28} at initial outpatient presentation (time point T1) and one year later (time point T3), with difference values, effect strengths, and the results of an analysis of variance

	T1 mean (SD)	T3 mean (SD)	T3-T1 mean (SD)	Pre/post ES
Adaptive disorders (F43)	20.20 (3.87)	21.31 (3.24)	1.11 (3.27)	0.31
Simple disorder of activity and attention (F90.0, 90.8, 90.9)	19.62 (3.65)	20.61 (3.55)	0.989 (3.54)	0.27
Hyperkinetic conduct disorder (F90.1)	17.53 (4.04)	19.14 (4.06)	1.61 (3.87)	0.39
Conduct disorder (F91, F92)	19.06 (3.82)	20.60 (3.19)	1.54 (3.43)	0.43
Specific emotional disorder of childhood (F93)	19.55 (3.85)	21.26 (3.49)	1.71 (3.39)	0.46
Other diagnoses	19.39 (3.86)	20.71 (3.55)	1.31 (3.49)	0.35
		F	df	P value
Results of analysis of variance with repeated measurement (factor 1, diagnosis; factor 2, time point)	Time point	73.44	1/541	<0.0005
	Diagnosis	6.186	4/541	<0.0005
	Time point $\times$ diagnosis	4.076	4/541	0.367 (n.s.)

We can, therefore, conclude the following:

- Mentally ill children and adolescents have a markedly poorer quality of life than the general population, as we pointed out in an earlier publication (Mattejat and Remschmidt 2006).
- Children with ADHD have a more severe impairment of the quality of life than children with specific emotional disorders and conduct disorders,.
- The improvement in quality of life after one year of treatment seems to be somewhat less marked in patients with ADHD than in patients with specific emotional disorders and conduct disorders, although this finding does not reach the level of statistical significance.

On the basis of these findings, we can now ask the more specific question how such results come about, and in particular, which aspects of the quality of life are most severely impaired in patients with ADHD. This issue can be explored with the aid of findings from the ILK individual ratings.

Individual ratings (dichotomized)

The frequencies (in percent) of problematic ILK ratings are indicated in Tables 11 and 12. To create these tables, we dichotomized the individual ratings in such a way (for the procedure, cf. the ILK handbook, Mattejat and Remschmidt 2006) as to make it apparent whether a rating indicating potential problems was given for any particular aspect of the quality of life. Thus, for example, the value “51.1” in the cell at top left in Table 11 indicates that 51.1% of the patients with adaptive disorders had problems in school.

Looking through Table 11, one sees that the patients with ADHD had more severe difficulties in school than patients in the other groups at the time of first presentation; the same was true in the area designated “Alone,” i.e., in their ability to occupy themselves when alone and to pursue their own interests. In particular, patients with hyperkinetic conduct disorders (but not those with simple ADHD) were also markedly impaired with respect to social contact within the family and with other children. All of these differences were statistically significant.

As seen in Table 12, considerable improvement was found one year after presentation. The following aspects are particularly noteworthy with respect to the patients with ADHD:

- Patients with simple ADHD experienced marked improvement in school: at the end of one year of treatment, their ratings in this area were comparable to those of the patients in the other diagnostic groups.

- This improvement was not seen among patients with hyperkinetic conduct disorders. These patients had the worst ratings of all not only in the scholastic area but also in the areas “Family,” “Other children,” and “Alone.”
- Finally, a point that merits particular emphasis is that the parents of children with ADHD gave themselves the least favorable ratings of the parents in any group with respect to the stress they experienced as a result of their child’s problems (Item 10). This was true both on presentation and at the end of a year of treatment.

Discussion

The first thing we can conclude from these findings is that the test-metrical qualities of the instrument we devised (the ILK) render it usable and capable of holding its own among other, comparable instruments (Mattejat and Remschmidt 2006; Riley et al. 2006b). Our study has three special properties: First, it was carried out on an unselected group of patients who presented consecutively, without gaps, to 9 outpatient practices for child and adolescent psychiatry. In these practices, the patients were treated under the normal conditions of everyday life (“treatment as usual”). Second, our study enables a comparison of the quality-of-life scores of patients with ADHD with those of children and adolescents with other mental illnesses. Third, we report on changes in the quality of life over a one-year period (longitudinal observation).

Interestingly, the patients in the different diagnostic groups did not differ from one another to any significant extent, at least as perceived by their parents, with respect to changes in their psychiatric problems (Table 8) or changes in their overall well-being (Table 9). Nonetheless, improvement was noted at one year in about 80% of the patients, though admittedly less commonly in the ADHD group than in the other diagnostic groups. The quality-of-life scores of these children and adolescents in psychiatric care were substantially lower than those of children in the general population; this corroborates findings reported by us in an earlier publication (Mattejat and Remschmidt 2006). Looking at the longitudinal changes in quality-of-life scores for all of the individual diagnostic groups, we find significant differences, not just between one time point and another but also between one diagnosis and another (see Table 10). The patients in the two ADHD groups (simple activity and attention disorder and hyperkinetic conduct disorder) had the lowest quality-of-life scores, both on initial presentation and one year later. In general, the overall sample experienced a highly statistically significant improvement at the end of one year. An analysis of

Table 11 Frequency of problematic individual ratings, in percent, on initial ambulatory presentation (time point T1)

Diagnostic groups	Aspects of the quality of life: ILK individual ratings								
	School	Family	Other children	Alone	Health	Nerves/ mood	All together	Stress, child	Stress, parents
	1	2	3	4	5	6	7	8	10
Adaptive disorders (F43)	51.1	14.0	27.4	30.2	9.4	50.0	31.9	63.2	77.9
Simple disorder of activity and attention (F90.0, 90.8, 90.9)	57.9	17.9	30.6	51.0	7.7	52.9	28.4	58.8	80.1
Hyperkinetic conduct disorder (F90.1)	57.1	35.5	58.7	51.6	14.1	69.8	44.4	60.3	79.7
Conduct disorder (F91, F92)	40.0	25.4	41.8	43.3	7.5	65.6	38.8	60.6	80.0
Specific emotional disorder of childhood (F93)	42.6	13.9	31.7	39.3	10.6	59.8	32.0	56.3	70.2
Other diagnoses	50.9	19.3	34.9	43.9	9.4	57.5	33.0	59.4	77.4
Differences between diagnostic groups: <i>P</i> value derived from the Pearson χ^2 test	0.025	0.003	0.000	0.008	0.591	0.045	0.149	0.890	0.306

Table 12 Frequency of problematic individual ratings, in percent, one year after the initial ambulatory presentation (time point T3)

Diagnostic groups	Aspects of the quality of life: ILK individual ratings										
	School	Family	Other children	Alone	Health	Nerves/ mood	All together	Stress, child	Stress, child, from Tx	Stress, parents	Stress, parents, from Tx
	1	2	3	4	5	6	7	8	9	10	11
9Adaptive disorders (F43)	36.6	15.6	20.8	21.9	5.2	29.5	11.5	38.9	32.3	31.2	34.7
Simple disorder of activity and attention (F90.0, 90.8, 90.9)	35.1	14.9	19.8	43.4	7.1	40.9	16.3	35.4	25.5	47.7	33.3
Hyperkinetic conduct disorder (F90.1)	46.9	32.8	38.1	43.8	9.7	51.6	23.4	40.6	25.0	54.7	29.7
Conduct disorder (F91, F92)	33.3	17.9	17.9	36.4	4.5	42.4	16.4	28.4	29.7	35.8	28.8
Specific emotional disorder of childhood (F93)	35.2	7.3	21.3	26.8	11.4	36.1	14.8	28.3	30.6	32.2	29.3
Other diagnoses	36.5	15.8	22.2	35.0	7.7	39.3	16.0	34.2	28.3	40.7	31.7
Differences between diagnostic groups: <i>P</i> value derived from the Pearson χ^2 test	0.476	0.000	0.030	0.001	0.333	0.064	0.369	0.277	0.695	0.002	0.851

variance revealed no significant interaction between time point and diagnosis, i.e., the improvement occurred to a comparable extent across all diagnostic groups. Although patients with ADHD, when compared to those in other groups (e.g., patients with specific emotional disorders and conduct disorders), were found to have an especially poor quality of life at the outset, they did show a certain degree of improvement after a year of treatment, which, however, failed to reach the level of statistical significance. When we look beyond the global quality-of-life scores to consider each of the individual aspects of the quality of life that are assessed by the ILK instrument, we find, predictably, that children with simple activity and attention disorders and those with hyperkinetic conduct disorders had the most severe problems in school (see Table 11). Patients with hyperkinetic conduct disorders were also markedly impaired in the two domains designated “Other children” and “Alone,” both on initial presentation and one year

later. Nonetheless, the patients in the two ADHD groups experienced marked improvement at one year in domains 1 through 4 of the ILK, as a comparison of Table 11 with Table 12 reveals. Finally, we point out that the parents of ADHD children experienced considerably more stress than the parents of children with any other diagnosis; this is particularly evident from the relevant ILK ratings after one year of treatment (see Table 12).

Our study confirms that the parents of children with ADHD suffer a great deal of stress (Harpin 2005), but also that therapeutic interventions can improve the quality of life (Matza et al. 2004). Longitudinal studies over longer periods of observation are still lacking, but our study, carried out over a time span of one year, clearly provides grounds for optimism.

Limitations: In this article, we have been able to describe no more than a part of our extensive project. Alongside standardized telephone interviews and the ILK

(Inventory for the Assessment of Life Quality in Children and Adolescents), we also used many other instruments, including the Marburg Symptom Scales (Mattejat and Remschmidt 2010), standardized basic documentation, the Child Behavior Checklist, the Questionnaire for the Evaluation of Treatments (*Fragebogen zur Beurteilung von Behandlungen*, FBB), and interviews of various kinds. The data obtained with these instruments cannot be discussed here, and this fact limits, to some extent, the informative value of the present article. A further limitation is that the data presented here were collected by many different people in a total of 9 outpatient practices for child and adolescent psychiatry. On the other hand, this might also be construed as a positive feature, as long as a high degree of agreement exists among informants. This statement is permissible and is additionally supported by the fact that all of the instruments used were standardized, and all of them have relatively good test-metric properties. Finally, we have only presented parents' judgments in this article, rather than the judgments of the patients themselves, or of their psychotherapists. We intend to present these in future publications.

Acknowledgments This study was performed in collaboration with the child and adolescent psychiatric practices of Dr. Edwin Fischer, Dr. Dagmar Hoehne, Dr. Kerstin Köhl, Dr. Le Lam, Dr. Martin Neuhaus, Dr. Klaus-Ulrich Oehler, Dr. Christa Schaff, Dr. Oya Uzelli-Schwarz, and Dr. Dipl.-Psych. Franz Wienand. We thank these colleagues for their outstanding cooperation.

References

- Bachmann M, Bachmann CJ, John K, Heinzel-Gutenbrunner M, Remschmidt H, Mattejat F (2010) The effectiveness of child and adolescent psychiatric treatments in a naturalistic outpatient setting. *World Psychiatry* 9(2):111–117
- Bullinger M, Von Mackensen S, Kirchberger I (1994) KINDL - ein Fragebogen zur Erfassung der gesundheitsbezogenen Lebensqualität von Kindern. *Zeitschrift für Gesundheitspsychologie* 2(1):64–67
- Deutsche Gesellschaft für Kinder- und Jugendpsychiatrie, Bundesarbeitsgemeinschaft Leitender Klinikärzte für Kinder- und Jugendpsychiatrie, Psychosomatik und Psychotherapie und dem Berufsverband der Ärzte für Kinder- und Jugendpsychiatrie, Psychosomatik und Psychotherapie (2007) In: Leitlinien zu Diagnostik und Therapie von psychischen Störungen im Säuglings-, Kindes- und Jugendalter.vol 3. Auflage Deutscher Ärzteverlag, Köln
- Dilling H, Mombour W, Schmidt MH, Schulte-Markwort E (Hrsg.) (1997) Internationale Klassifikation psychischer Störungen (ICD-10) der WHO, Huber, Bern
- Harpin VA (2005) The effect of ADHD on the life of an individual, their family, and community from preschool to adult life. *Arch Dis Child* 90(Suppl 1):2–7
- Mattejat F, Remschmidt H (2006) ILK Inventar zur Erfassung der Lebensqualität bei Kindern und Jugendlichen. Huber, Bern
- Mattejat F, Remschmidt H (2010) MSR marburger symptom rating. Huber Verlag, Bern
- Mattejat F, Trosse M, John K, Bachmann M, Remschmidt H (2006) kjp-Qualität. Modell-Forschungsprojekt zur Qualität ambulanter kinder- und jugendpsychiatrischer Behandlungen. Abschlussbericht, Juli 2006. Görlich & Weiershäuser, Marburg
- Matza LS, Rentz AM, Secnik K, Swensen AR, Revicki DA, Michelson D, Spencer T, Newcorn JH, Kratochvil CJ (2004) The link between health-related quality of life and clinical symptoms among children with attention-deficit hyperactivity disorder. *J Devel Behav Pediatr* 25:166–174
- Riley AW, Coghill D, Forrest C-B, Lorenzo M-J, Ralston S-J, Spiel G, ADORE study group (2006a) Validity of the health-related quality of life assessment in the ADORE study: parent report form of the CHIP-child edition. *Eur Child Adolesc Psychiatry* 15(Suppl 1):I/63–I/71
- Riley AW, Spiel G, Coghill D, Döpfner M, Fallisard B, Lorenzo M-J, Preuss U, Ralston S-J, the ADORE study group (2006b) Factors related to health-related quality of life (HRQoL) among children with ADHD in Europe at entry into treatment. *Eur Child Adolesc Psychiatry* 15(Suppl 1):I38–I45
- Riley AW, Lyman LM, Spiel G, Döpfner M, Lorenzo M-J, Ralston S-J, the ADORE study group (2006c) Reliability, validity, and factor structure of a brief questionnaire for families of children with ADHD. *Eur Child Adolesc Psychiatry* 15(Suppl 1):72
- Rothenberger A, Coghill D, Döpfner M, Falissard B, Steinhausen HC (2006) ADORE (Attention-deficit/hyperactivity disorder observational research in Europe). Baseline data of a large-scale longitudinal study. *Eur Child Adolesc Psychiatry* 15:78
- Stiensmeier-Pelster J, Schürmann M, Duda K (2000) Depressions-Inventar für Kinder und Jugendliche (DIKJ). Handanweisung (2.überarb. u. neunorm. Aufl.) Göttingen, Hogrefe
- Swensen AR, Birnbaum HG, Secnick K, Marynchenko M, Greenberg P, Claxton A (2003) Attention-deficit/hyperactivity disorder: increased costs for patients and their families. *J Am Acad Child Adolesc Psychiatry* 42:1415–1423
- WHOQOL Group (1993) Study protocol for the World Health Organization project to develop a Quality of Life assessment instrument (the WHOQOL). *Qual Life Res* 2:153–159
- WHOQOL Group (1995) The world health organization quality of life assessment (WHOQOL): position paper from the world health organization. *Soc Sci Med* 41:1403–1409